

AI-enhanced OTC hearing aids

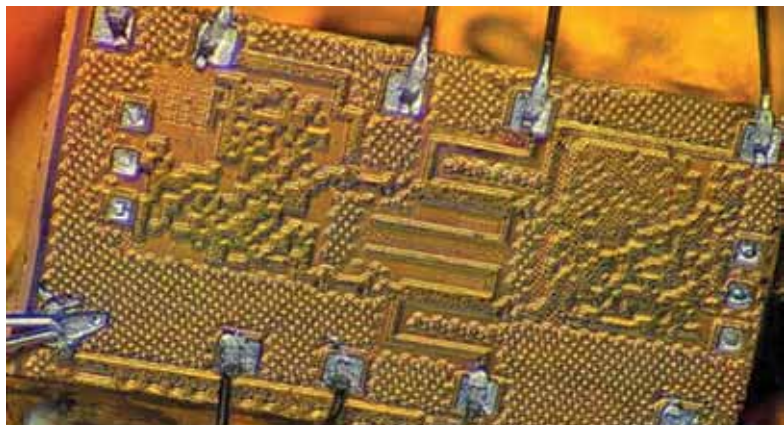


Ambiq has announced a breakthrough in OTC (over the counter) hearing aids by integrating AI-powered speech enhancement technology. OTC hearing aids do not require implantation or other surgical intervention. This innovation advances hearing-assist devices, addressing challenges with real-time neural network processing. With the Apollo510 and neuralSPOT AI Deep Learning solutions, OTC hearing aids provide speech clarity and noise reduction through on-device audio processing. Key technological features include the neural network speech de-noiser (NNSD), isolating speech by removing background noise, and neural network acoustic environment detection (NNAED), which adjusts audio to match surroundings. This power-efficient design extends battery life, eliminates costly ASICs, and accelerates product cycles.

Ambiq revolutionises OTC hearing aids with AI-powered speech enhancement

AI-driven design of advanced wireless chips

Researchers at Princeton Engineering and the Indian Institute of Technology have developed an AI-driven approach to revolutionise wireless chip design, drastically reducing costs and time. AI generates complex electromagnetic structures and circuits based on design parameters, completing weeks-long tasks in hours. This method enables unconventional designs with greater energy efficiency and broader frequency range capabilities, surpassing traditional algorithms. The system can create intricate broadband amplifiers and aims to design entire wireless chips, which is crucial for applications like autonomous driving, radar, and gesture recognition. Human oversight remains essential to address potential flaws in AI-generated configurations despite its advancements.



The AI design features unusual and efficient circuitry patterns (Credit: Emir Ali Karahan, Princeton University)